

Part 1 (40 marks)

Q1) True or False? (6 marks)

- a) **False.** PHP types are dynamic. No declaration needed.
- b) **False.** XML is for data structure & transport, not display.
- c) **True.**
- d) **False.** `insertBefore` is called on the parent node, not always `document`.
- e) **True.** This is called type juggling.
- f) **False.** Ajax can use other formats like JSON or plain text.

Q2) Fill in the blanks. (8 marks)

- a) `length`
- b) `returnValue`
- c) `XMLHttpRequest`
- d) `database`
- e) `$_GET`
- f) `WHERE`
- g) `valid`
- h) `integer, float`

Q3) XML vs. HTML differences. (4 marks)

- **Purpose:** HTML displays data (web pages). XML transports/stores data.
- **Syntax:** HTML has preset tags. XML has custom tags. XML is stricter (case-sensitive, must close tags).

Q4) Client-Server Model. (3 marks)

1. **Client (browser)** requests a resource via URL.
2. **Server** gets the request, finds the resource.
3. **Server** sends it back. Browser shows it.

Q5) `$_GET` vs. `$_POST`. (2 marks)

- `$_GET`: Data sent in the URL. Visible. Size limit.
- `$_POST`: Data sent in request body. Hidden. No real size limit.

Q6) `include` vs. `require`. (2 marks)

- **Purpose:** Both insert a PHP file's content into another. For code reuse.
- **Error Handling:** If file not found, `include` gives a warning and continues. `require` gives a fatal error and stops.

Q7) SQL Query. (3 marks)

```
SELECT productName, price
FROM products
WHERE category = 'Electronics' AND price > 500;
```

Q8) PHP Script Output. (4 marks)

Output: Final Result: 198

Why: The script loops through [1, 2, 3, 4, 5, 6]. It starts with `$result = 1`, adds odd numbers, and multiplies by even numbers.

1+1=2 -> 2*2=4 -> 4+3=7 -> 7*4=28 -> 28+5=33 -> 33*6=198.

Q9) PHP Price Calculation Function. (8 marks)

```
<?php
function calculateFinalPrice($initialPrice, $userStatus, $membershipYears) {
    $discount = 0.0;

    // Set base discount
    if ($userStatus === 'vip') {
        $discount = 0.20;
    } elseif ($userStatus === 'member') {
        $discount = 0.10;
    }

    // Add extra discount for loyal members
    if (($userStatus === 'vip' || $userStatus === 'member') && $membershipYears >=
5) {
        $discount += 0.05;
    }

    // Return final price
    return $initialPrice * (1 - $discount);
}
?>
```